

### **A Beer Game in Nephrology: The Peritoneal Dialysis Solution Supply Chain**

**The scenario.** Consider the chain that keeps one home peritoneal dialysis patient in therapy. Each month that patient receives dialysate bags, transfer sets, and cyclor cassettes. Four parties stand between the sterile-fluid plant and the closet in the patient's house: the dialysis organization's home program, which places patient-level orders; a regional distributor that fills them; the manufacturer's distribution center; and the plant itself. Each party sees only the orders placed immediately downstream, each holds protective inventory, and each is measured on its own fill rate. That is the Beer Game's structure exactly — a serial chain, local information, local incentives.

**The disturbance.** A hurricane closes the plant for three weeks and the manufacturer moves to allocation. Home programs hear this before any shortage reaches them, and each does the locally sensible thing: it orders two to three months of supply per patient instead of one. The distributor's order book triples while patient census has not moved at all. The distribution center reads that spike as demand growth and orders aggressively upstream. The plant schedules overtime against what now looks structural. By the time that production arrives the programs are sitting on months of inventory and stop ordering, so the plant reads a collapse and cuts the line — shortly before the buffer empties and real orders return.

**Where the delays live.** Two delays drive the oscillation, and they compound. The information delay is the lag between a program changing its order quantity and anyone upstream reading that change as demand rather than noise, which takes weeks of order history. The material delay is physical: compounding, sterilization, quality release, and transit run six to ten weeks. Because the material delay exceeds the ordering interval, every party orders again before its previous order arrives. Each shortage is ordered against two or three times, and the chain over-corrects by construction.

**The communication breakdown.** Nobody lies, and each party reports accurately within its own frame. The home program never tells the distributor that its census is flat and it is deliberately building buffer, because no channel exists for saying so. The distributor has no

visibility into census at all; it sees order quantities and nothing else. True demand — patients performing a fixed number of exchanges a day, a figure that barely moves — is the one quantity no node in the chain observes. Orders stand in for consumption, and under stress that proxy stops tracking what it represents.

**Why the consequences exceed the Beer Game's.** In Senge's version the penalty is carrying cost and backlog. Here it is clinical. Patients under allocation are short-shipped, and some stretch supply by shortening dwell times or skipping an exchange, degrading ultrafiltration and clearance. Others transfer to in-center hemodialysis, and a meaningful share never return home. A three-week plant outage converts into permanent modality change for real people. That outcome is invisible to every node: the plant sees tons shipped, the distributor sees fill rate, and the only party positioned to see technique failure is the clinic, which has no view of the supply dynamics that caused it.

**The cause is structural.** Senge's central claim is that the same structure produces the same behavior regardless of who fills the roles, and this chain demonstrates it. The reflexive explanations — the manufacturer failed, the programs hoarded — are both the enemy-is-out-there response, and both miss that every actor behaved rationally on the information available. Replace every person in the chain with a more careful one and the oscillation returns.

**What would change it.** Because the cause is structural, exhortation and better forecasting will not help; only changes to information flow will. Share true consumption rather than order quantities, so upstream parties see census and exchange counts instead of a proxy. Allocate on census rather than order history, which removes the incentive to inflate. And shorten the information delay by making consumption visible continuously rather than monthly. That last remedy is nearly free for a remote monitoring system: the cyclor already reports every exchange it runs, so consumption is measured nightly and simply is not routed to the people who need it. The data exists; the feedback loop does not.